Metadata standards and controlled vocabularies

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What is a metadata standard

- How does a standard make data FAIR?
- What are the ingredients for defining a standard?
- Where to find metadata standards?
- Which tools can be used in connection with standards?

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Metadata standards and “Findable”, “Accessible”

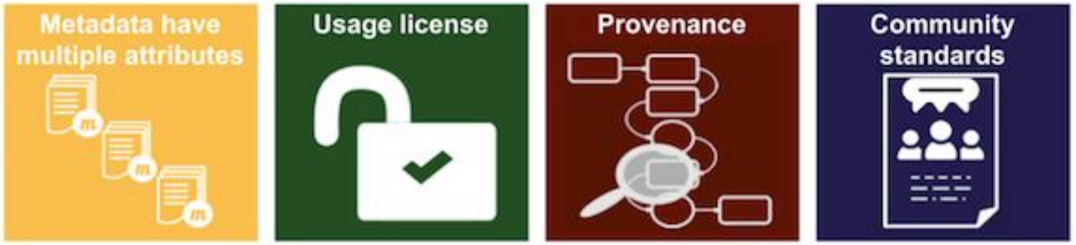
- The metadata requires a persistent identifier
- Metadata standard compliant with repositories checklist
- Metadata fields used to identify/retrieve the data

- Persistency of metadata (also when the data is not available)

Metadata standards and “Interoperable”

- Interoperability fully relies on metadata
- Vocabularies and ontologies ensure that standards are FAIR
- A well-defined standard can be linked with standards describing other type of data
- Focus on machine-actionability

Metadata standards and “Reusable”



- Richer metadata fields enhance reusability
- Metadata should describe data provenance
- Everything needs to follow community standards
  - Alignment with repository

ArrayExpress — MINSEQE; MIAME

PRIDE Archive (PRoteomics IDEntifications Database) — HUPO-PSI ; TraML; MIAPE

ENA (European Nucleotide Archive) — SRA-XML

Example: ENA sample checklists 1

There is a minimum amount of information required during ENA sample registration and all samples must conform to a defined checklist of expected metadata values. The most suitable checklist depends on sample type.

https://www.ebi.ac.uk/ena/browser/checklists

Example: ENA sample checklists 2

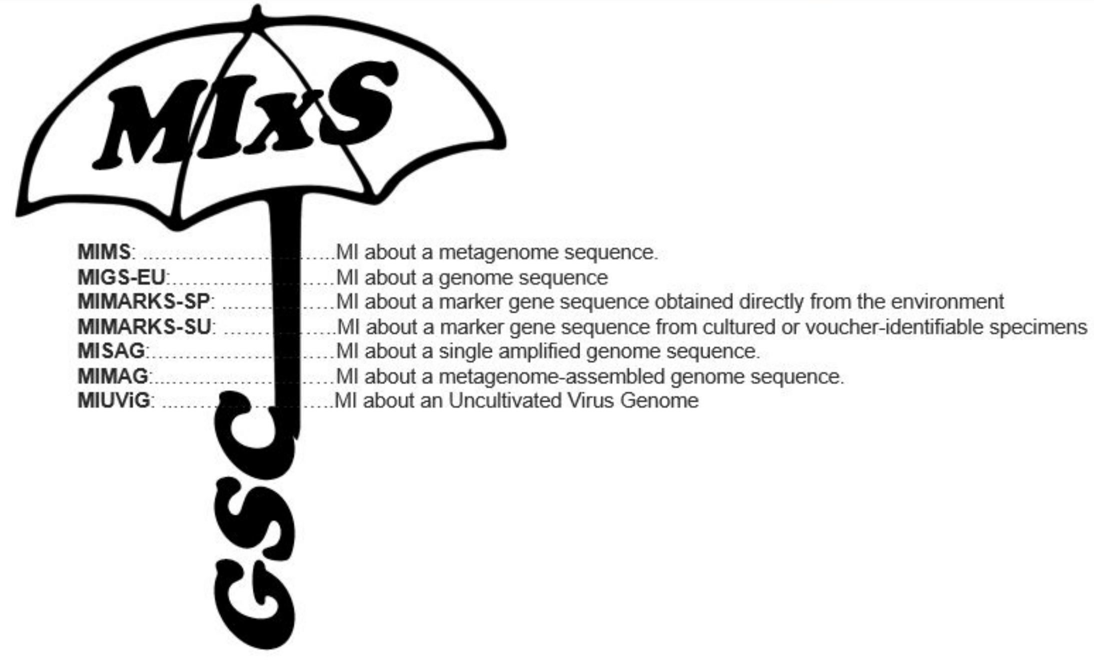
ENA European Nucleotide Archive

Sample Checklists

Accession	Name
ERC000012	GSC MIxS air
ERC000013	GSC MIxS host associated
ERC000014	GSC MIxS human associated

https://www.ebi.ac.uk/ena/browser/checklists

genomic STANDARDS consortium



https://www.gensc.org/pages/standards-intro.html

Example: ENA sample checklists 3

European Nucleotide Archive

ERC000014	**GSC MIxS human associated**

Field Name	Field Format	(Field Restriction)
project name	free text	
experimental factor	free text	
ploidy	free text	
number of replicons	restricted text	regular expression

https://www.ebi.ac.uk/ena/browser/checklists

Example: ENA sample checklists 4

European Nucleotide Archive

ERC000014 GSC MIxS human associated

number of replicons	restricted text	regular expression	
smoker	text choice	ex-smoker	non-smoker	smoker

https://www.ebi.ac.uk/ena/browser/view/ERC000014

Controlled vocabularies

Lists of pre-selected answers for metadata fields are known as controlled vocabularies.
They are useful to avoid situations such the one below, where things got out of control

How many ways can you say "female"?

18-day pregnant females	female (lactating)	individual female	worker caste (female)
2 yr old female	female (pregnant)	lgb*cc females	sex: female
400 yr. old female	female (outbred)	mare	female, other
adult female	female parent	female (worker)	female child
asexual female	female plant	monosex female	femal
castrate female	female with eggs	ovigerous female	3 female
cf.female	female worker	oviparous sexual females	female (phenotype)
cystocarpic female	female, 6-8 weeks old	worker bee	female mice
dikaryon	female, virgin	female enriched	female, spayed
dioecious female	female, worker	pseudohermaphroditic female	femlale
diploid female	female(gynoecious)	remale	metafemale

Controlled vocabularies – when not used

How many ways can you say “female”?

18-day pregnant females
2 yr old female
400 yr. old female
adult female
asexual female
castrate female
cf.female
cystocarpic female
dikaryon
dioecious female
diploid female
f
famale
femail
female
female - worker
female (alate sexual)
female (calf)
hen

female (lactating)
female (pregnant)
female (outbred)
female parent
female plant
female with eggs
female worker
female, 6-8 weeks old
female, virgin
female, worker
female(gynoecious)
femele
female, pooled
femalen
females
females only
gynoecious
healthy female
probably female (based on morphology)

individual female
lgb*cc females
mare
female (worker)
monosex female
ovigerous female
oviparous sexual females
worker bee
female enriched
pseudohermaprhoditic female
remale
semi-engorged female
sexual oviparous female
sterile female worker
strictly female
tetraploid female
thelytoky
female (gynoecious)

worker caste (female)
sex: female
female, other
female child
femal
3 female
female (phenotype)
female mice
female, spayed
femlale
metafemale
sterile female
normal female
sf
vitellogenic replete female
worker
hexaploid female
female (f-o)

female (note: this sample was originally provided as a \"male\" sample to us and therefore labeled this way in the brawand et al. paper and original geo submission; however, detailed data analyses carried out in the meantime clearly show that this sample stems from a female individual)"

Courtesy of N. Silvester, European Nucleotide Archive, EMBL-EBI

Ontologies 1

While controlled vocabularies offer more control over free text boxes, this solution is not ideal as different data sources could use different options with e.g. a different granularity.

The way to address this is to replace words in this list with persistent identifiers provided by an external sources. This would enable cross-linking between data resources i.e. interoperability

Ontologies 2

Granularity is very important to find the right terminology. Using a well-defined hierarchy usually helps finding the right information.

- quality (1,879)
  - physical object quality (1,665)
    - organismal quality (215)
      - biological sex (35)
        - phenotypic sex (19)
          - female (2)
            - intact female
            - spayed female

Phenotype And Trait Ontology

An ontology of phenotypic qualities (properties, attributes or characteristics)

PID

http://purl.obolibrary.org/obo/PATO_0000383

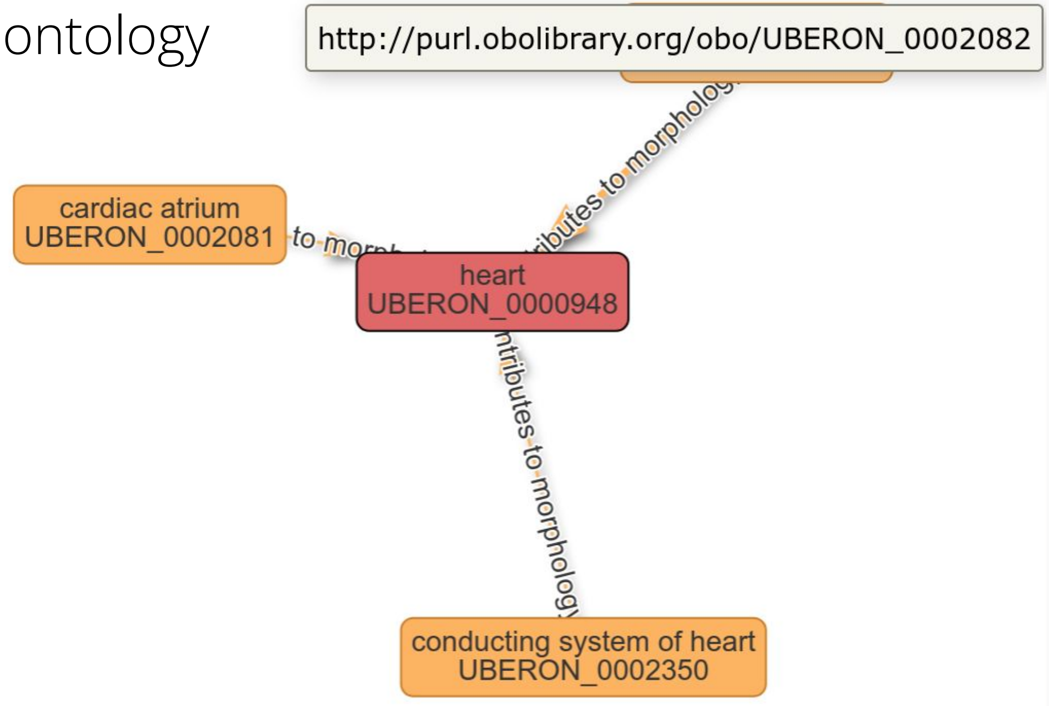
https://www.ebi.ac.uk/ols4/ontologies/pato/classes/http%253A%252F%252Fpurl.obolibrary.org%252Fobo%252FPATO_0000383

NORWAY

Ontologies 3

- On top of the **hierarchy**, formal relations between terms are also relevant to interlink data
- These can be visualised as **graphs**

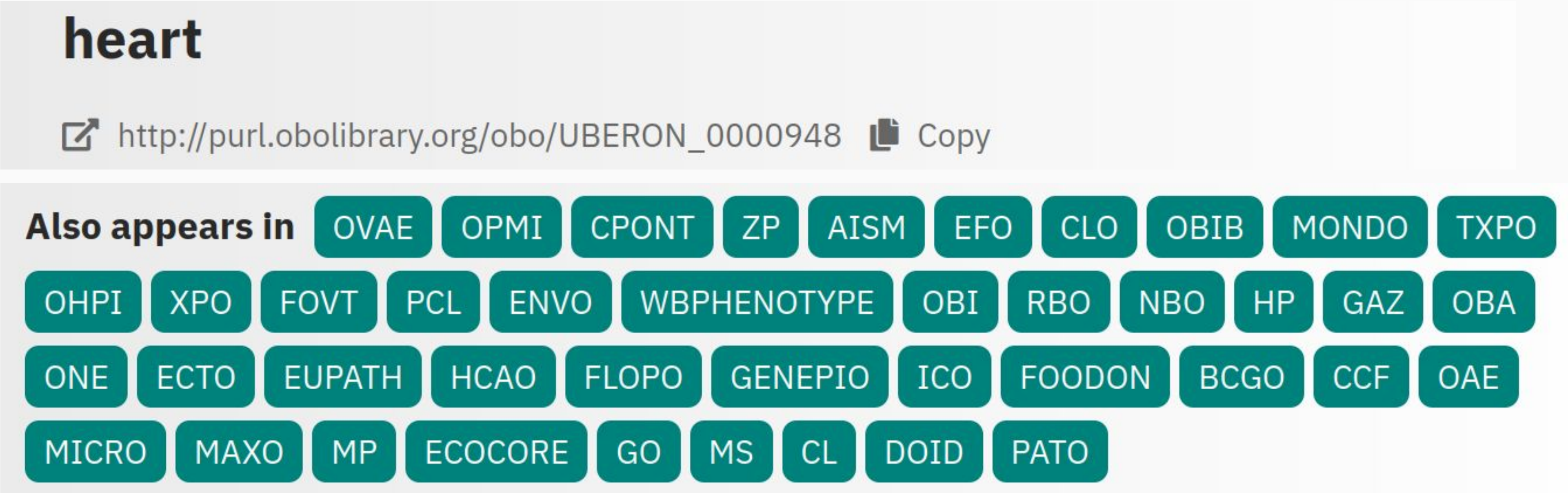
Uber-anatomy ontology



https://www.ebi.ac.uk/ols4/ontologies/pato/classes/http%253A%252F%252Fpurl.obolibrary.org%252Fobo%252FUBERON_0000948

Ontologies 4

Cross-linking of terms across ontologies allows for an improved standardisation of terms and definitions.



https://www.ebi.ac.uk/ols4/ontologies/uberon/classes/http%253A%252F%252Fpurl.obolibrary.org%252Fobo%252FUBERON_0000948

Ontologies 5

Definition and synonyms to facilitate search functionalities.

heart

http://purl.obolibrary.org/obo/UBERON_0000948 Copy

A myogenic muscular circulatory organ found in the vertebrate cardiovascular system composed of chambers of cardiac muscle. It is the primary circulatory organ.

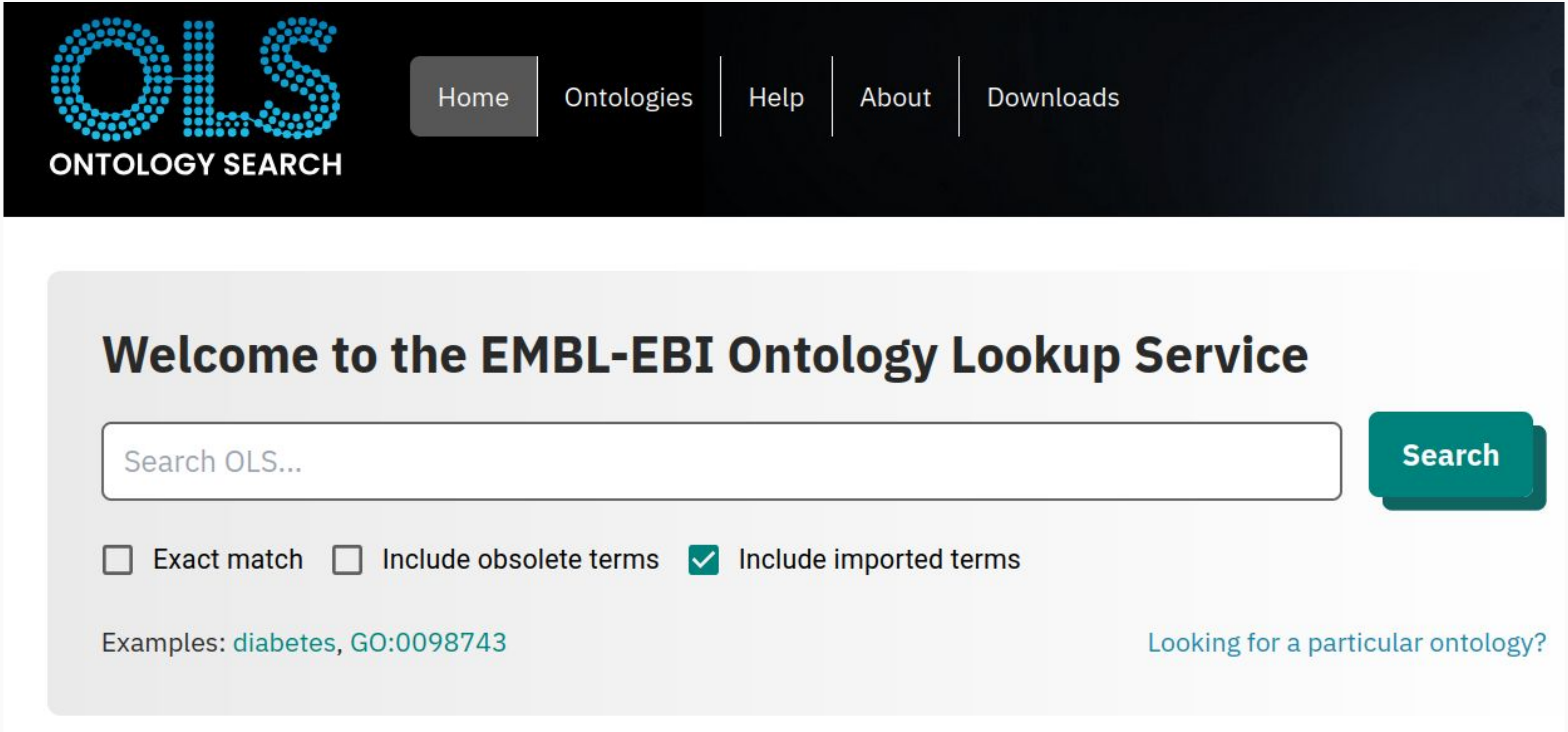
Synonym `branchial heart` `cardium` `chambered heart` `vertebrate heart`

Ontologies – summary

To summarise, an ontology is a set of concept and categories in a subject area that:

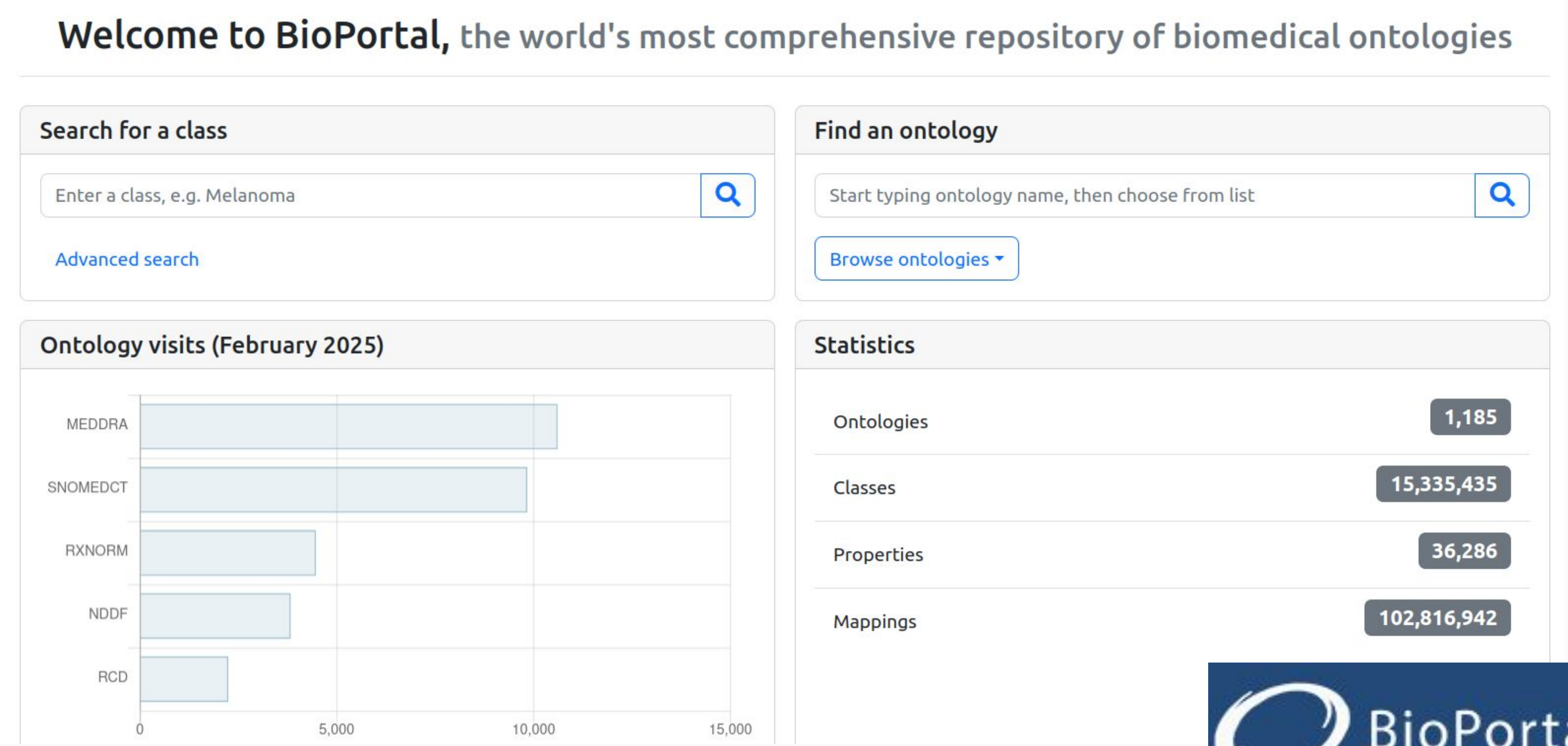
- Shows and define properties
- Shows the relation between these properties
- Provides a persistent identifier to these properties
- Follows a specific tree-like hierarchy
- Is cross-linked with other ontologies

Ontologies – where to find them 1



https://www.ebi.ac.uk/ols4/

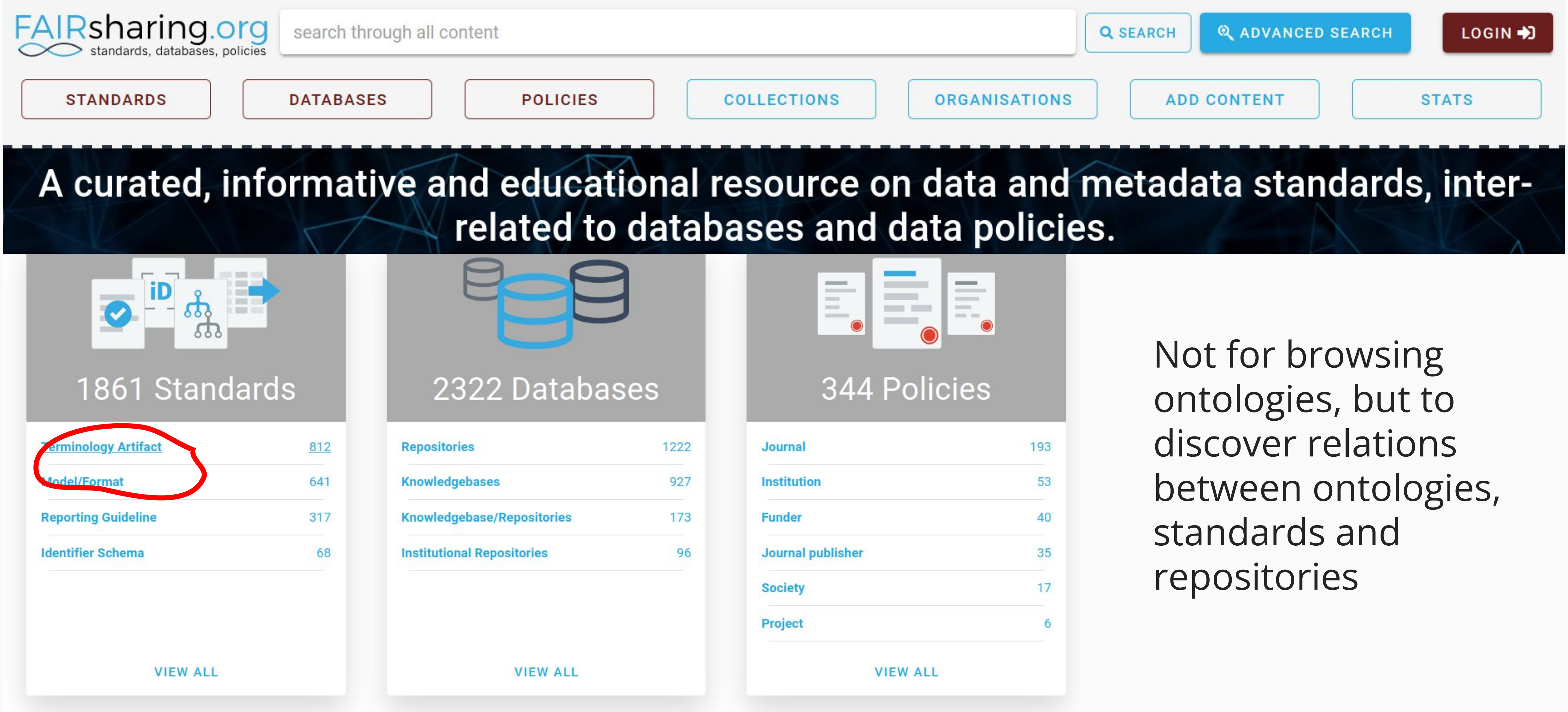
Ontologies – where to find them 2



BioPortal

https://bioportal.bioontology.org/

Ontologies - where to find them 3



Not for browsing ontologies, but to discover relations between ontologies, standards and repositories

https://fairsharing.org/search?recordType=terminology_artefact

Metadata tracking platforms

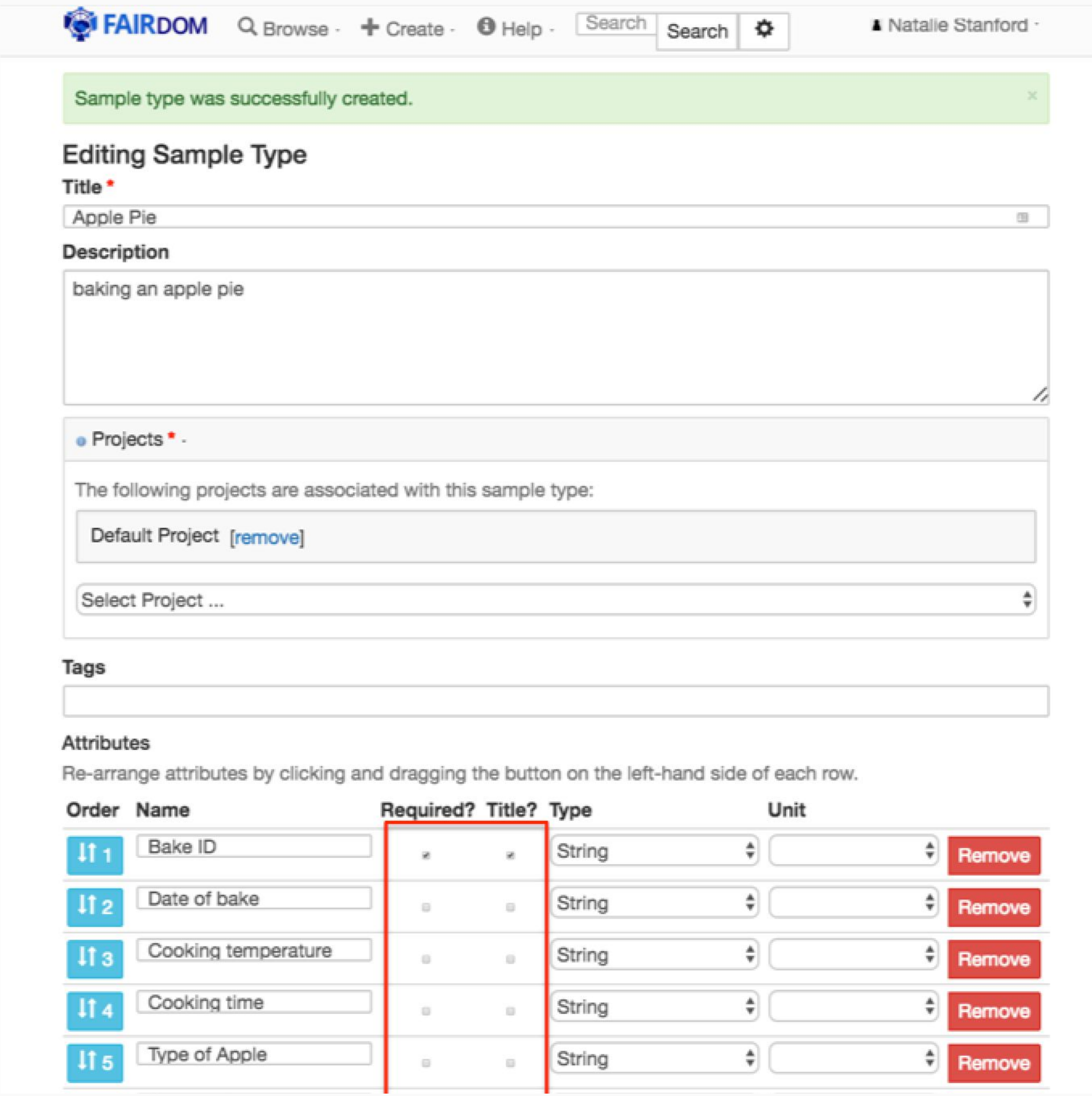
Domain specific:

- COPO for plant sciences
- MOLGENIS for biobanking
- Omero for imaging data

Customisable (domain expertise required)

- Proprietary ELNs/LIMS -
  - often poor support for ontologies
- openBIS - open source ELN/LIMS
- FAIRDOM SEEK

https://copo-project.org/
https://www.molgenis.org/
https://openbis.ch/
https://seek4science.org/



FAIRDOM SEEK

The SEEK platform is a web-based resource for sharing heterogeneous scientific research datasets, models or simulations, processes and research outcomes. It preserves associations between them, along with information about the people and organisations involved.

- National users (via Digital Life):

1000100 1010011
DIGITAL SALMON

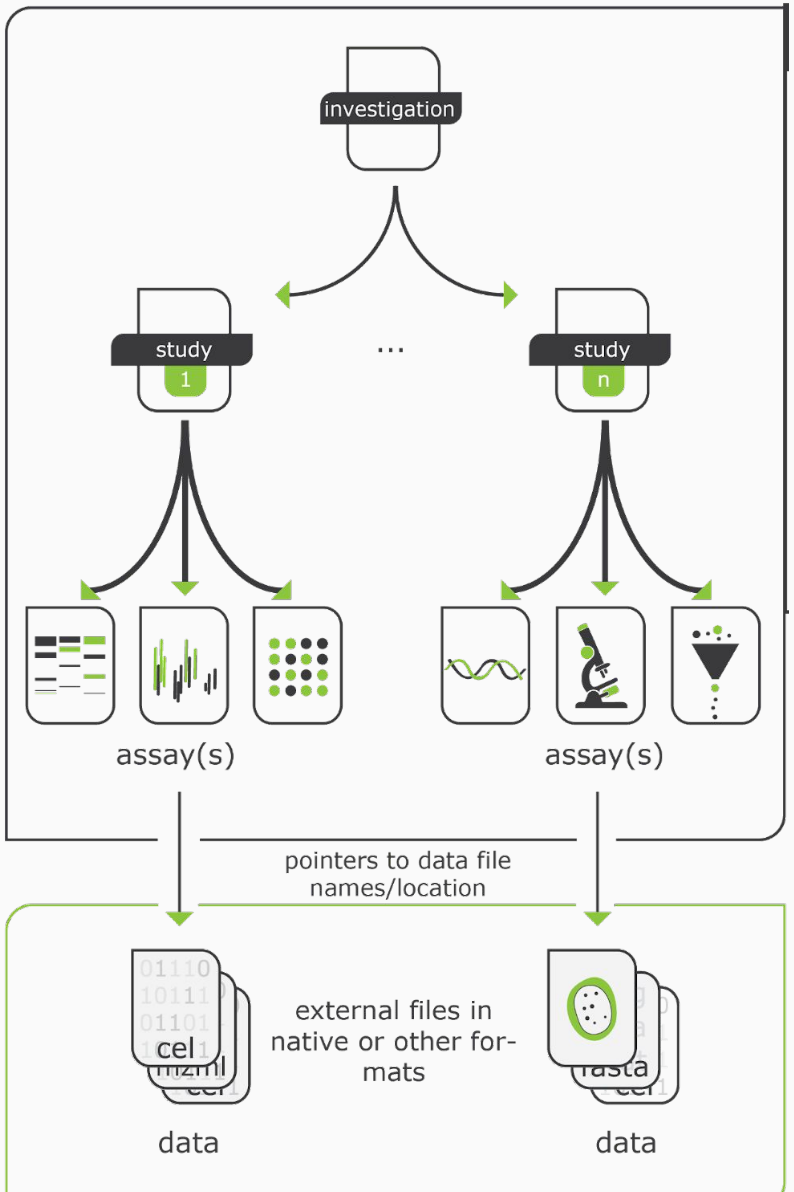
dCod 1.0
decoding the systems toxicology of atlantic cod

- Target new users (via ELIXIR)

nor seq

NAPI

The ISA model



- Investigation
  - Persons
  - Organizations
  - Publications
- Study(s)
  - Design
  - Factor
  - Protocol
- Assay(s)
  - Measurement
  - Technology
  - Materials
  - Data

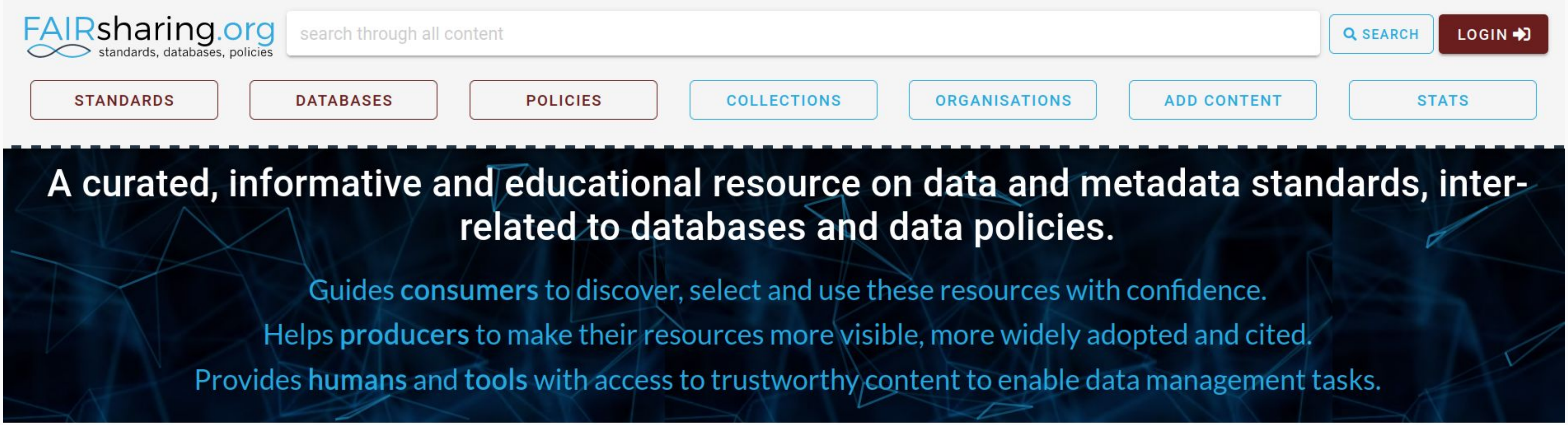
Domain-agnostic standard with typical metadata fields for a project in the life sciences

Possibility of semantic annotations

elixir Recommended Interoperability Resources

Standards and repositories - where to find them

FAIRsharing.org
standards, databases, policies



https://fairsharing.org/

Thank you

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